

FCS PRACTICE SESSION 1 · PROBLEMS 9–26

Degree Sequences, Forests, Connectivity and Counting

Complete worked solutions to the remaining eighteen problems, and the starred one

19 problems · Full proofs and constructions · All results machine-verified

This is the companion volume to *Degrees, Regularity, Isomorphism and Connectivity*, which solves problems 1–8 of the same sheet. Here we treat problems 9–26 together with the starred problem that sits between 20 and 21. Nothing from the first volume is assumed beyond the handshake lemma and the definition of a component, both restated below, so this document can be read on its own.

Every solution is obtained by graph-theoretic means. Several of the problems are not posed in the language of graphs — a game played with pencil and paper (12), a question about a table of numbers (23), a party (25), a tiled rectangle (26) — and in each the first step is to identify the graph hiding inside. Where a problem asks for a construction or an exhaustive list, the objects are exhibited and checked one by one rather than merely asserted to exist.

All computational claims were additionally confirmed by exhaustive machine search: every graph on at most six or seven vertices where that suffices, the complete game tree for problem 12, brute-force minimisation for the two extremal quantities of problem 14, and randomised tilings for problem 26. Those checks are reported alongside the arguments; they corroborate the proofs rather than replace them.

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★ three colours, 123 edges — at most 42 vertices · **21** acyclic orientations · **22** three edges, three cuts · **23** deleting a column · **24** colouring K_4 · **25** 13 people · **26** tiling a rectangle

What is assumed

Graphs are finite; they are simple unless the problem explicitly allows otherwise. We write $n = |V(G)|$, $m = |E(G)|$, $d(v)$ for the degree of v , $\Delta(G)$ and $\delta(G)$ for the largest and smallest degree, and $c(G)$ for the number of connected components. K_n , P_n and C_n are the complete graph, the path and the cycle on n vertices, and $\overline{G}$ is the complement of G . Four facts are used throughout.

Lemma A (handshake lemma) $\sum_{v \in V(G)} d(v) = 2m$.

Proof. Count the pairs (v, e) with v an endvertex of e in two ways: by edges, each contributing 2; by vertices, vertex v contributing $d(v)$. $\square$

Lemma B (components are degree-closed) If H is a component of G and $x \in V(H)$, then every neighbour of x lies in H ; hence $d_H(x) = d_G(x)$ and $|V(H)| \geq d_G(x) + 1$.

Proof. A neighbour y of x is joined to x by an edge, so it lies in the same component. The $d_G(x)$ neighbours together with x itself are $d_G(x) + 1$ distinct vertices of H . $\square$

Lemma C (edge addition, EAL) Adding one edge e to G either creates no cycle through e , in which case $c(G + e) = c(G) - 1$, or creates a cycle through e , in which case $c(G + e) = c(G)$. In particular one edge lowers the number of components by at most one.

Lemma D (forests) A forest on n vertices with c components has exactly $n - c$ edges. Consequently a forest has at most $n - 1$ edges, with equality exactly for trees.

Proof. Build the forest edge by edge from the edgeless graph, which has n components. No added edge ever lies on a cycle, so by Lemma C each addition lowers the component count by exactly one. After m additions the count is $n - m$, and this equals c . $\square$

PROBLEM 9 Graphs with all degrees distinct

pigeonhole

PROBLEM

Determine all finite simple graphs that do not have two vertices of the same degree.

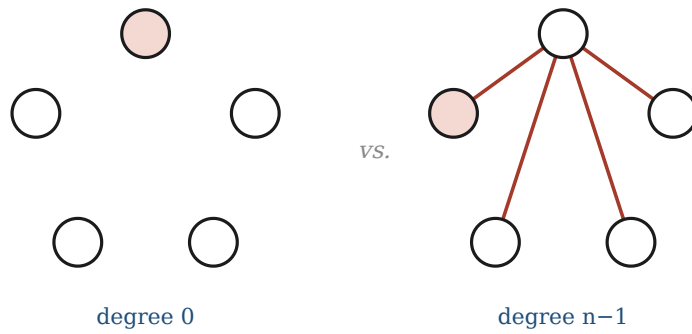
Theorem 9.1 The only finite simple graph whose degrees are pairwise distinct is the one-vertex graph K_1 .

Proof. For $n = 1$ the condition is vacuous, and K_1 qualifies. So let G be simple with $n \geq 2$ vertices and suppose its n degrees are pairwise distinct.

In a simple graph a vertex has at most $n - 1$ neighbours, so every degree lies in the n -element set $\{0, 1, \dots, n - 1\}$. We have n pairwise distinct degrees drawn from a set of size n , so every value in $\{0, 1, \dots, n - 1\}$ is taken, each exactly once. In particular there is a vertex u with $d(u) = 0$ and a vertex w with $d(w) = n - 1$, and $u \neq w$ because $0 \neq n - 1$ (here $n \geq 2$).

Now $d(w) = n - 1$ means w is adjacent to all $n - 1$ other vertices, u among them. But then $d(u) \geq 1$, contradicting $d(u) = 0$.

So no simple graph on $n \geq 2$ vertices has pairwise distinct degrees. $\square$



one vertex isolated, another joined to everyone — including the isolated one

The obstruction: distinct degrees on $n \geq 2$ vertices would force both an isolated vertex and a vertex joined to everything — and the second must be joined to the first.

ANSWER

Only K_1 . For $n \geq 2$ the n distinct degrees would have to exhaust $\{0, 1, \dots, n-1\}$, producing simultaneously a vertex of degree 0 and a vertex of degree $n-1$ — which is impossible, since the latter is adjacent to the former.

The usual formulation Turned around, the theorem says: *every simple graph on at least two vertices has two vertices of equal degree*. This is the standard "two people at a party have the same number of acquaintances" puzzle. Note where simplicity is used: with loops or parallel edges the degrees are no longer confined to $\{0, \dots, n-1\}$ and the argument collapses — two vertices joined by one edge and by a second parallel edge attached to a loop can easily have different degrees.

Machine check Every simple graph on $n \leq 6$ vertices was enumerated: exactly one has pairwise distinct degrees, namely K_1 at $n = 1$; for $n = 2, \dots, 6$ there are none.

PROBLEM 10 The degree sequence 2, 2, 2, 4, 5, 5 *forced adjacencies*

PROBLEM

Prove that if the degree sequence of a 6-vertex graph is 2, 2, 2, 4, 5, 5, then it is not simple.

The two cheap tests are passed, so they cannot be the obstruction: the entries sum to $2 + 2 + 2 + 4 + 5 + 5 = 20$, which is even (so $m = 10$), and the largest entry 5 does not exceed $n - 1 = 5$. The obstruction is structural, and the two vertices of degree 5 are what force it.

Theorem 10.1 *No simple graph on six vertices has degree sequence 2, 2, 2, 4, 5, 5.*

Proof. Suppose G were such a simple graph, on the vertex set $\{a, b, c, d, x, y\}$ with

$$d(a) = d(b) = d(c) = 2, \quad d(d) = 4, \quad d(x) = d(y) = 5.$$

Step 1: x and y are joined to everything. Since G is simple on six vertices, a vertex has at most 5 neighbours, and they are distinct. So $d(x) = 5$ forces x to be adjacent to each of a, b, c, d, y , and likewise y to each of a, b, c, d, x .

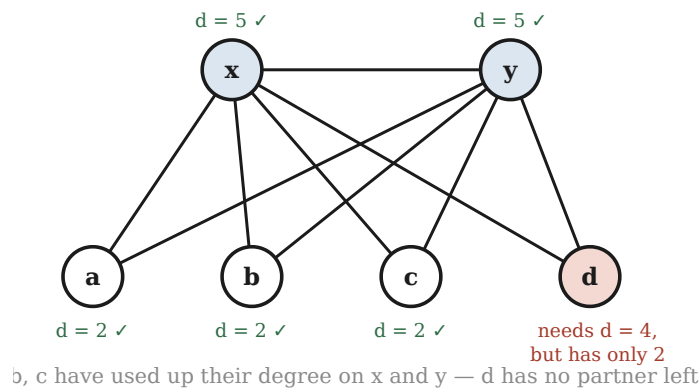
Step 2: a, b and c are used up. By Step 1 each of a, b, c is already adjacent to both x and y , so each already has degree 2. As their prescribed degree is exactly 2, none of them is adjacent to anything else. In particular

$$ad, bd, cd \notin E(G).$$

Step 3: d cannot reach degree 4. By Step 1, d is adjacent to x and to y ; by Step 2 it is adjacent to none of a, b, c . Since G is simple, d cannot be joined to x or y a second time, and there is no loop at d . Hence $d(d) = 2$, contradicting $d(d) = 4$.

The assumption of simplicity is what fails, so no simple graph has this degree sequence.

□



The adjacencies forced by $d(x) = d(y) = 5$. Once a, b, c have spent their degree 2 on x and y , the vertex d is stranded at degree 2.

The same conclusion from Havel-Hakimi

The reduction of Havel and Hakimi deletes the largest entry d_1 and subtracts 1 from each of the next d_1 entries; the original sequence is graphical if and only if the reduced one is. Applied here:

round	current sequence	operation	reduced sequence
1	(5, 5, 4, 2, 2, 2)	delete the leading 5; subtract 1 from the next 5 entries	(4, 3, 1, 1, 1)
2	(4, 3, 1, 1, 1)	delete the leading 4; subtract 1 from the next 4 entries	(2, 0, 0, 0)
3	(2)	the leading 2 needs 2 further vertices, but only 0 remain	not graphical ✗

Table 10.1 — the Havel-Hakimi reduction of 5, 5, 4, 2, 2, 2 breaks down, confirming that the sequence is not graphical.

At the third step the leading entry 2 demands two further vertices of positive residual degree, and none remain. So the sequence is not graphical — in agreement with Theorem 10.1.

ANSWER

No simple graph has this degree sequence, so any graph realising $2, 2, 2, 4, 5, 5$ must contain a loop or a pair of parallel edges. The reason is that the two vertices of degree $5 = n - 1$ are adjacent to everything; that already exhausts the degree of the three vertices of degree 2, leaving the vertex of prescribed degree 4 with only x and y available — degree 2.

The sequence *is* realisable if parallel edges are allowed Simplicity is exactly what fails, and nothing more. Taking the edges

$$xy, xd, xd, xa, xb, yd, yd, yc, ya, bc$$

(with xd and yd each doubled) gives a loopless multigraph on a, b, c, d, x, y with ten edges and degrees $d(x) = d(y) = 5$, $d(d) = 4$, $d(a) = d(b) = d(c) = 2$, as required.

A third route: Erdős-Gallai The Erdős-Gallai criterion says a non-increasing sequence with even sum is graphical exactly when $\sum_{i \leq k} d_i \leq k(k-1) + \sum_{i > k} \min(d_i, k)$ for every k . For $5, 5, 4, 2, 2, 2$ the inequality holds at $k = 1, 2, 4, 5, 6$ but fails at $k = 3$:

$$5 + 5 + 4 = 14 \quad \text{versus} \quad 3 \cdot 2 + \min(2, 3) + \min(2, 3) + \min(2, 3) = 6 + 6 = 12.$$

The three vertices of largest degree want 14 edge-endpoints, but they can absorb at most 6 among themselves and at most $3 \cdot 2 = 6$ from the rest.

Machine check All $2^{15} = 32\,768$ simple graphs on six labelled vertices were enumerated; none has degree sequence $2, 2, 2, 4, 5, 5$. The multigraph above was checked to have the right degrees, and the Erdős-Gallai inequalities were evaluated for all six values of k , failing exactly at $k = 3$.

PROBLEM 11 Eleven vertices and forty-five edges

averaging

PROBLEM

Show that if $G = (V, E)$ satisfies $|V| = 11$ and $|E| = 45$, then it has a vertex of degree at least 9.

Theorem 11.1 Every graph with 11 vertices and 45 edges has a vertex of degree at least 9.

Proof. By the handshake lemma (Lemma A),

$$\sum_{v \in V} d(v) = 2|E| = 2 \cdot 45 = 90.$$

Suppose, for contradiction, that every vertex had degree at most 8. Summing this bound over the 11 vertices gives

$$90 = \sum_{v \in V} d(v) \leq 11 \cdot 8 = 88,$$

which is false. Hence some vertex has degree at least 9. $\square$

Equivalently: the average degree is $90/11 = 8.\overline{18}$, and the maximum of a finite list is at least its average, so $\Delta(G) \geq \lceil 90/11 \rceil = 9$.

ANSWER

The degrees sum to $2 \cdot 45 = 90$. If all eleven were at most 8 the sum could not exceed $11 \cdot 8 = 88 < 90$. So **some vertex has degree at least 9**.

Two remarks The argument uses nothing about simplicity, so it holds for multigraphs as well. And the bound 9 cannot be improved: since $\binom{11}{2} = 55$, deleting the ten edges of a Hamiltonian path from K_{11} leaves a simple graph with 11 vertices and exactly 45 edges in which every degree is 10 minus the path degree, i.e. 8 or 9. So $\Delta = 9$ is attained and the conclusion "degree at least 10" would be false.

PROBLEM 12**The cycle-avoiding game***the game length is an invariant***PROBLEM**

Alice and Bob play the following game. Alternately they draw edges between n given vertices. The player who first obtains a cycle loses the game. Who wins if both players play an optimal strategy, the one who moves first, or the other one?

The surprise is that *neither player has a choice that matters*. Whatever they do, the game lasts exactly $n - 1$ safe moves and the n -th move loses; so the winner is decided by the parity of n before a single edge is drawn.

Theorem 12.1 *For $n \geq 2$ the first player wins if n is even, and the second player wins if n is odd.*

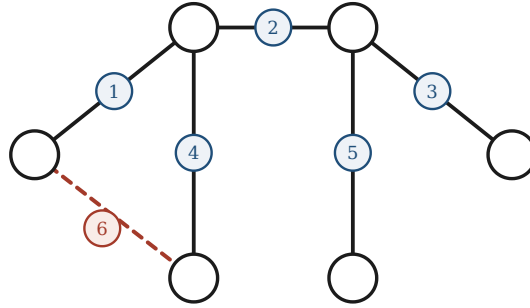
Proof. At every moment the drawn edges form a graph F on the n vertices, and as long as nobody has lost, F is *acyclic* — a forest. We make two observations.

(i) *A safe move exists precisely while $|E(F)| < n - 1$.* By Lemma D a forest on n vertices with c components has $n - c$ edges. So $|E(F)| < n - 1$ means $c \geq 2$: there are two different components, and joining a vertex of one to a vertex of the other creates no cycle (a cycle through the new edge would give a second connection between the two components). That move is safe. Conversely if $|E(F)| = n - 1$ then $c = 1$, so F is a spanning tree, and *every* further edge uv closes a cycle, namely the new edge together with the unique u - v path in F .

(ii) *The game therefore lasts exactly $n - 1$ safe moves.* A player who draws a cycle loses at once, so under optimal play nobody does so voluntarily; and by (i) a safe move is available as long as fewer than $n - 1$ edges are down. After $n - 1$ safe moves the picture is a spanning tree, and by (i) the player to move must create a cycle. There is such an edge to draw, since $\binom{n}{2} - (n - 1) = \binom{n-1}{2} \geq 1$ for $n \geq 3$, so that player moves and loses. (For $n = 2$ the single edge is already drawn and the second player has no move at all; with the usual convention that a player unable to move loses, the count below is unchanged.)

(iii) *Parity decides.* Moves 1, 3, 5, ... belong to the first player and moves 2, 4, 6, ... to the second. The losing move is move number n . So the first player loses exactly when n is odd, and the second player loses exactly when n is even.

□



6: five safe moves build a spanning tree; move 6 must close a cycle

$n = 6$: the numbered edges are the five forced safe moves, in whatever order and shape the players choose. The dashed sixth edge — any remaining edge — must close a cycle.

ANSWER

The first player wins when n is even; the second player wins when n is odd.

Neither player can influence the outcome. A drawn position is always a forest, a safe move exists exactly while the forest has fewer than $n - 1$ edges, and any maximal forest is a spanning tree with exactly $n - 1$ edges. So the game always consists of $n - 1$ safe moves followed by a losing n -th move, and move n belongs to the first player precisely when n is odd.

Machine check The complete game tree was searched for $n = 2, \dots, 7$, with both players playing optimally. The winner is the first player for $n = 2, 4, 6$ and the second player for $n = 3, 5, 7$ — exactly the parity rule.

PROBLEM 13 One hundred vertices, one of degree 66 component sizes

PROBLEM

Let us assume that simple graph G has 100 vertices and each has degree at least 33, furthermore, there is a vertex of G that is incident with at least 66 edges. Prove that G is connected. (Midterm '15)

Theorem 13.1 Let G be a simple graph on 100 vertices with $\delta(G) \geq 33$, and suppose some vertex v_0 satisfies $d(v_0) \geq 66$. Then G is connected.

Proof. Suppose G were disconnected, and let H_0 be the component containing v_0 and H_1 any other component; they are vertex-disjoint.

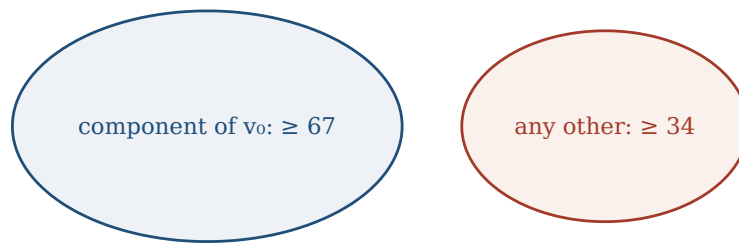
By Lemma B a component containing a vertex of degree d has at least $d + 1$ vertices — that vertex together with its d neighbours, all of which lie in the same component. Applying this twice:

$$|V(H_0)| \geq d(v_0) + 1 \geq 67, \quad |V(H_1)| \geq \delta(G) + 1 \geq 34.$$

Since H_0 and H_1 are disjoint subsets of $V(G)$,

$$100 = |V(G)| \geq |V(H_0)| + |V(H_1)| \geq 67 + 34 = 101,$$

a contradiction. Hence G has only one component. □



a second component would push the total past 100 vertices

Why a second component cannot fit: the component of v_0 already occupies at least 67 of the 100 vertices, and any further component needs 34 more.

ANSWER

G is **connected**. A component containing a vertex of degree d has at least $d + 1$ vertices. So the component of v_0 has at least 67 vertices and every other component at least $\delta + 1 = 34$; two of them would need $67 + 34 = 101 > 100$ vertices.

Both hypotheses are needed Neither alone suffices. With $\delta \geq 33$ but no high-degree vertex, two disjoint copies of K_{50} give a disconnected graph of minimum degree $49 \geq 33$. With a vertex of degree 66 but no lower bound elsewhere, a star $K_{1,66}$ together with 33 isolated vertices is disconnected. It is the combination $67 + 34 > 100$ that does the work — and it is tight: had the hypothesis read "degree at least 65", the numbers would give $66 + 34 = 100$, and indeed $K_{66} \dot{\cup} K_{34}$ is a disconnected graph with $\delta = 33$ and a vertex of degree 65.

Machine check The scaled-down analogue — $n = 7$, $\delta \geq 2$, some vertex of degree ≥ 4 , where $5 + 3 = 8 > 7$ plays the role of $67 + 34 > 100$ — was verified over all graphs on seven vertices: every graph satisfying the hypothesis is connected.

PROBLEM 14

Ten deletions and thirty-three additions

cycle rank and component count

PROBLEM

How many edges does the 66-vertex graph G have that satisfies the following two properties? (a) At least 10 edges must be deleted from G in order to obtain a cycle-free graph. (b) At least 33 edges must be added to G in order to obtain a connected graph.

Read the two conditions as they are meant: 10 is the *minimum* number of deletions that makes G acyclic, and 33 the *minimum* number of additions that makes it connected. Both minima have closed formulas in n , m and $c = c(G)$, and together they pin m down.

Theorem 14.1 For any graph G with n vertices, m edges and c components:

- (i) the minimum number of edges whose deletion makes G acyclic is $m - n + c$;
- (ii) the minimum number of edges whose addition makes G connected is $c - 1$.

Proof of (i). Choose a spanning forest $F \subseteq G$ — a spanning tree in each component. By Lemma D it has $n - c$ edges, so deleting the other $m - (n - c)$ edges leaves the acyclic graph F : that many deletions suffice. Conversely, suppose deleting a set D leaves $G - D$ acyclic. Deleting edges never merges components, so $c(G - D) \geq c$, and by Lemma D the forest $G - D$ has $n - c(G - D) \leq n - c$ edges. Hence $|D| = m - |E(G - D)| \geq m - (n - c)$. $\square$

Proof of (ii). By Lemma C each added edge lowers the number of components by at most one, so getting from c components to 1 needs at least $c - 1$ additions. They also suffice: pick one vertex from each component and join them consecutively by $c - 1$ new edges. $\square$

Solving for m

Here $n = 66$. Condition (b) and Theorem 14.1(ii) give

$$c - 1 = 33 \quad \implies \quad c = 34.$$

Condition (a) and Theorem 14.1(i) then give

$$m - n + c = 10 \quad \implies \quad m = n - c + 10 = 66 - 34 + 10 = \boxed{42}.$$

ANSWER

G has **42 edges**. The minimum number of deletions making a graph acyclic is $m - n + c$ and the minimum number of additions making it connected is $c - 1$. So $c - 1 = 33$ gives $c = 34$, and $m - 66 + 34 = 10$ gives $m = 42$.

Such a graph exists Take a path on 33 vertices (32 edges), add ten chords from its first vertex, and throw in 33 isolated vertices. That is 66 vertices, 42 edges and 34 components; its spanning forest has $66 - 34 = 32$ edges, so $42 - 32 = 10$ deletions are needed to make it acyclic, and $34 - 1 = 33$ additions to connect it. Both were confirmed by brute-force minimisation.

A word on the wording Taken hyper-literally, "at least 10 must be deleted" and "at least 33 must be added" would only say $m - n + c \geq 10$ and $c - 1 \geq 33$, which do not determine m — for instance 66 vertices in 34 components with a K_{33} among them satisfies both and has 528 edges. The question asks for *the* number of edges, so the two numbers are the exact minima, as used above.

Machine check Both formulas of Theorem 14.1 were checked against brute-force minimisation on 120 random graphs with up to seven vertices, and the witness graph above was verified to have 66 vertices, 42 edges, 34 components, deletion number 10 and addition number 33.

PROBLEM 15 All 5-vertex graphs with 2, 8 and 3 edges

exhaustive lists

PROBLEM

Find (up to isomorphism) all n -vertex m -edge graphs for a) $n = 5, m = 2$; b) $n = 5, m = 8$; c) $n = 5, m = 3$.

Throughout, isolated vertices are part of the graph — all five vertices are always present. A convenient way to organise the search is by the *edge structure*: discard the isolated vertices, list the possible graphs with the prescribed number of edges and no isolated vertex, and then pad back up to five vertices whenever there is room.

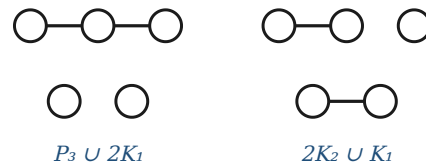
a) $n = 5, m = 2$

Two edges either share an endvertex or they do not, and there is no third possibility.

- They share a vertex: the two edges form a path P_3 on three vertices, leaving two isolated vertices — the graph $P_3 \dot{\cup} 2K_1$, with degree sequence $2, 1, 1, 0, 0$.
- They are disjoint: they occupy four vertices, leaving one isolated — the graph $2K_2 \dot{\cup} K_1$, with degree sequence $1, 1, 1, 1, 0$.

Both fit inside five vertices, and their degree sequences differ, so they are not isomorphic.

There are exactly two.



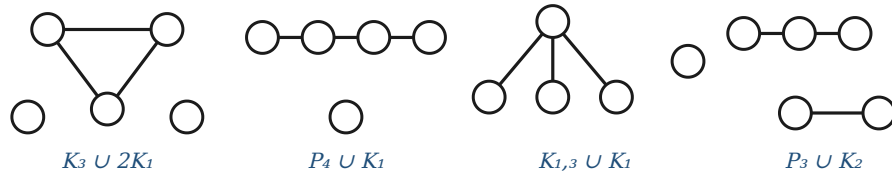
The two graphs with $n = 5, m = 2$.

c) $n = 5, m = 3$

We take (c) before (b), because (b) will be read off from (a). Sort the three edges by how their components split.

- *All three edges in one component.* A connected graph with three edges has at most four vertices. On three vertices it must be the triangle K_3 ; on four vertices it has $4 - 1 = 3$ edges, so it is a tree, and the trees on four vertices are the path P_4 and the star $K_{1,3}$. This gives $K_3 \dot{\cup} 2K_1$ (degrees $2, 2, 2, 0, 0$), $P_4 \dot{\cup} K_1$ (degrees $2, 2, 1, 1, 0$) and $K_{1,3} \dot{\cup} K_1$ (degrees $3, 1, 1, 1, 0$).
- *Split $2 + 1$.* The two-edge part must be P_3 (three vertices, as disjoint edges would already use four) and the single edge K_2 two more: five vertices exactly, giving $P_3 \dot{\cup} K_2$ with degrees $2, 1, 1, 1, 1$.
- *Split $1 + 1 + 1$.* Three pairwise disjoint edges need six vertices — impossible here.

The four degree sequences $2, 2, 2, 0, 0$; $2, 2, 1, 1, 0$; $3, 1, 1, 1, 0$; $2, 1, 1, 1, 1$ are pairwise different, so these four graphs are pairwise non-isomorphic. **There are exactly four.**

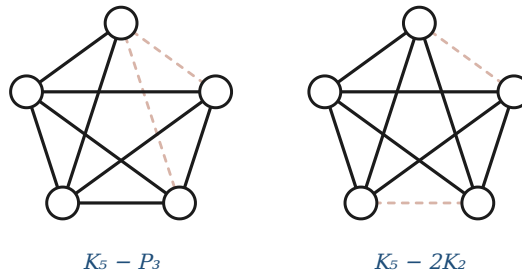
The four graphs with $n = 5$, $m = 3$.**b) $n = 5$, $m = 8$**

Here the complement does the work. On five vertices there are $\binom{5}{2} = 10$ possible edges, so $|E(\overline{G})| = 10 - 8 = 2$. Complementation is an involution and preserves isomorphism in both directions ($G_1 \cong G_2 \iff \overline{G_1} \cong \overline{G_2}$, because a bijection preserves adjacency exactly when it preserves non-adjacency). Hence

$$G \mapsto \overline{G}$$

is a bijection between the 5-vertex graphs with 8 edges and those with 2 edges, taken up to isomorphism. By part (a) there are **exactly two**, namely the complements of the two graphs found there:

- $\overline{P_3 \dot{\cup} 2K_1} = K_5 - P_3$: remove two adjacent edges from K_5 ; degrees 4, 4, 3, 3, 2.
- $\overline{2K_2 \dot{\cup} K_1} = K_5 - 2K_2$: remove two disjoint edges from K_5 ; degrees 4, 3, 3, 3, 3.



dashed = the two removed edges

The two graphs with $n = 5$, $m = 8$, drawn as K_5 with two edges removed.**ANSWER**

a) two graphs — $P_3 \dot{\cup} 2K_1$ and $2K_2 \dot{\cup} K_1$.

b) two graphs — $K_5 - P_3$ and $K_5 - 2K_2$, the complements of those in (a).

c) four graphs — $K_3 \dot{\cup} 2K_1$, $P_4 \dot{\cup} K_1$, $K_{1,3} \dot{\cup} K_1$ and $P_3 \dot{\cup} K_2$.

Machine check All $2^{10} = 1024$ labelled graphs on five vertices were sorted into isomorphism classes. The number of classes with m edges is 1, 1, 2, 4, 6, 6, 6, 4, 2, 1, 1 for $m = 0, 1, \dots, 10$ — palindromic, as complementation requires. The entries at $m = 2, 3, 8$ are 2, 4, 2, and the classes found by machine are exactly the ones listed above.

PROBLEM 16**Graphs isomorphic to their complements***a parity obstruction***PROBLEM**

Are there 5-vertex graphs that are isomorphic to their complements? How about 6-vertex graphs?

A graph with $G \cong \overline{G}$ is called *self-complementary*. There is one immediate necessary condition.

Lemma 16.1 *If G is a self-complementary graph on n vertices then $\binom{n}{2}$ is even, i.e. $n \equiv 0$ or $1 \pmod{4}$.*

Proof. Every pair of distinct vertices is an edge of exactly one of G and $\overline{G}$, so

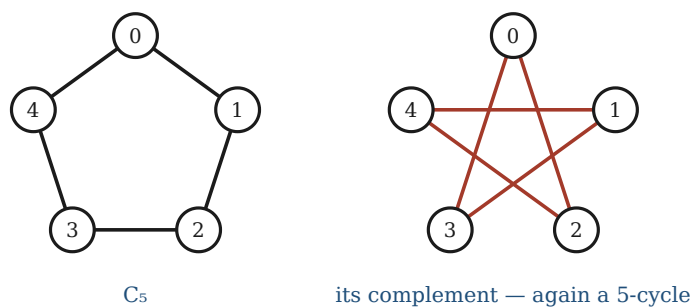
$|E(G)| + |E(\overline{G})| = \binom{n}{2}$. An isomorphism is a bijection on vertices carrying edges onto edges, so isomorphic graphs have equally many edges: $|E(G)| = |E(\overline{G})|$. Hence $\binom{n}{2} = 2|E(G)|$ is even. Since $\binom{n}{2} = n(n-1)/2$, evenness is exactly $n \equiv 0, 1 \pmod{4}$. $\square$

Five vertices: yes

Here $\binom{5}{2} = 10$ is even, so the obstruction does not apply — and indeed an example exists. Take the five-cycle C_5 on $\{0, 1, 2, 3, 4\}$ with edges $01, 12, 23, 34, 40$. Its complement has the five edges $02, 13, 24, 30, 41$, which form the closed walk

$$0 - 2 - 4 - 1 - 3 - 0,$$

a five-cycle again. Explicitly, $\varphi(i) = 2i \bmod 5$ is a bijection of $\{0, \dots, 4\}$ carrying the edges of C_5 onto the edges of $\overline{C_5}$: a difference of ± 1 is multiplied into ± 2 , and the edges of $\overline{C_5}$ are exactly the pairs at difference ± 2 . So $C_5 \cong \overline{C_5}$.



C_5 and its complement. The complement of a pentagon is the pentagram — which is again a pentagon, drawn differently.

Six vertices: no

Here $\binom{6}{2} = 15$ is odd, so by Lemma 16.1 there is no self-complementary graph on six vertices. Said directly: $|E(G)| = |E(\overline{G})|$ would force $15 = 2|E(G)|$, and 15 is not even.

ANSWER

Five vertices: yes. C_5 is isomorphic to its complement (which is again a 5-cycle); there are exactly two such graphs on five vertices.

Six vertices: no. A self-complementary graph must have $\binom{n}{2}$ even, and $\binom{6}{2} = 15$ is odd.

The second one, and what the condition does not give On five vertices there are exactly two self-complementary graphs: C_5 , and the *bull* — a triangle with a pendant edge attached to each of two of its vertices, degree sequence 3, 3, 2, 1, 1. Lemma 16.1 is only necessary, not sufficient: it permits $n = 4, 5, 8, 9, 12, 13, \dots$, and self-complementary graphs do exist for every such n (for $n = 4$, the path P_4), but the lemma alone does not prove that.

Machine check Complete enumeration: $n = 4$ has 1 self-complementary graph, $n = 5$ has 2 (C_5 and the bull), and $n = 6$ has 0.

PROBLEM 17 Connectivity needs $n - 1$ edges; *the edge addition lemma* n edges force a cycle

PROBLEM

Let G be an arbitrary n -vertex graph. Prove that if G is connected, then it has at least $n - 1$ edges. Also show that if it has n edges, then it contains a cycle.

Both halves follow from one bookkeeping principle: build G one edge at a time from the edgeless graph and watch the number of components fall.

Theorem 17.1 *Let G have n vertices, m edges and c components. Then $m \geq n - c$. In particular a connected graph has at least $n - 1$ edges.*

Proof. List the edges of G as $e_1, \dots, e_m$ and let G_k be the graph with all n vertices and the edges $e_1, \dots, e_k$; so G_0 is edgeless with $c(G_0) = n$, and $G_m = G$ with $c(G_m) = c$. By Lemma C each step lowers the number of components by at most one:

$$c(G_k) \geq c(G_{k-1}) - 1 \quad (k = 1, \dots, m).$$

Chaining these, $c = c(G_m) \geq c(G_0) - m = n - m$, that is $m \geq n - c$. For a connected graph $c = 1$ and $m \geq n - 1$. □

Theorem 17.2 *If an n -vertex graph has n edges, then it contains a cycle. More generally, any graph with $m \geq n$ edges contains a cycle.*

Proof. Suppose G had no cycle, i.e. G is a forest. Then by Lemma D it has exactly $n - c$ edges, where $c \geq 1$ is its number of components; hence

$$m = n - c \leq n - 1 < n,$$

contradicting $m \geq n$. So G contains a cycle. □

Lemma D itself is the same edge-by-edge count run under the extra assumption that no cycle ever appears: every one of the m additions then falls in the first case of the Edge Addition Lemma and drops the component count by exactly one, from n to $n - m$.

ANSWER

Add the edges one at a time. By the Edge Addition Lemma each new edge lowers the number of components by at most one, so starting from n components one needs at least $n - 1$ edges to reach a single component: **a connected n -vertex graph has at least $n - 1$ edges.**

If the graph were cycle-free, every addition would lower the count by *exactly* one, so a forest with $c \geq 1$ components has exactly $n - c \leq n - 1$ edges. **Hence n edges are impossible without a cycle.**

Both bounds are attained Trees show that $n - 1$ edges can suffice for connectivity, and a forest with c components shows $m = n - c$ is attainable in general. Together the two theorems say that the quantity $m - n + c \geq 0$ — the *cycle rank* — is zero exactly for forests. It reappears in problems 14 and 19 as the number of edges one must delete to destroy all cycles.

Machine check Both statements were verified over every graph on at most six vertices: no connected graph has fewer than $n - 1$ edges, and no graph with n or more edges is acyclic.

PROBLEM 18 **Add one edge, delete two***two competing budgets***PROBLEM**

Is it possible to obtain a connected graph from a graph of 88 vertices, 42 components and 111 edges by a sequence of steps, where one step consists of adding one edge and deleting two edges at the same time? (An edge deleted earlier can be added later.)

Two quantities are worth watching: the number of edges, which the step controls exactly, and the number of components, which it can only move slowly. They pull in opposite directions.

Theorem 18.1 *No such sequence of steps reaches a connected graph.*

Proof. Write $n = 88$, and let m_k and c_k be the number of edges and of components after k steps, so $m_0 = 111$ and $c_0 = 42$.

The edge budget. Each step adds one edge and deletes two, so $m_k = 111 - k$. By Theorem 17.1 a connected graph on 88 vertices needs at least 87 edges. So if the graph is connected after k steps then

$$111 - k \geq 87 \quad \implies \quad k \leq 24.$$

The component budget. Within one step, adding an edge lowers the number of components by at most one (Lemma C), while deleting edges can only raise it or leave it unchanged — deleting an edge never merges two components. Hence $c_k \geq c_{k-1} - 1$ for every k , regardless of the order in which the three edges are processed, and therefore

$$c_k \geq 42 - k.$$

Being connected means $c_k = 1$, which forces $1 \geq 42 - k$, i.e. $k \geq 41$.

The two requirements are incompatible: $41 \leq k \leq 24$ is empty. So no sequence of steps produces a connected graph. $\square$

ANSWER

No, it is impossible. Each step costs one edge net, so after k steps there are $111 - k$ edges; connectivity on 88 vertices needs 87 of them, giving $k \leq 24$. But each step can lower the number of components by at most one, so getting from 42 components down to 1 needs $k \geq 41$. Since $41 > 24$, the target is unreachable.

How close is it? Not close. Even spending the entire edge budget — 24 steps, ending at exactly 87 edges, the bare minimum for a spanning tree — removes at most 24 of the 41 components that must go. The obstruction is not a parity accident but a shortfall of 17 steps.

PROBLEM 19 Delete one edge, add two*the cycle rank as a monovariant***PROBLEM**

Is it possible to obtain a cycle-free simple graph from a 77-vertex simple graph of 66 edges and 42 components by applying a sequence of steps where a step consists of deletion of one edge and addition of two edges? (An edge deleted earlier is possible to be added back later.)

Now the edge count moves the other way, and the right quantity to track is the one that measures "how far from cycle-free" the graph is.

Definition The *cycle rank* of a graph is $\rho(G) = m - n + c$, where m, n, c are its numbers of edges, vertices and components. By Theorem 17.1 we have $\rho(G) \geq 0$, and by Lemma D, $\rho(G) = 0$ exactly when G is cycle-free.

Lemma 19.1 *Deleting one edge lowers ρ by at most 1; adding one edge never lowers ρ at all.*

Proof. Delete an edge e : then m drops by 1, and c either stays (if e lay on a cycle) or rises by 1 (if e was a bridge). So ρ changes by $-1 + 0 = -1$ or by $-1 + 1 = 0$. Add an edge e : by Lemma C, m rises by 1 and c either falls by 1 (no cycle through e) or stays (a cycle through e). So ρ changes by $+1 - 1 = 0$ or by $+1 - 0 = +1$. $\square$

Theorem 19.2 *No such sequence of steps reaches a cycle-free graph.*

Proof. At the start $n = 77$, $m_0 = 66$, $c_0 = 42$, so

$$\rho_0 = 66 - 77 + 42 = 31.$$

The rank budget. One step deletes one edge and adds two, so by Lemma 19.1 it lowers ρ by at most 1 in total. Reaching a cycle-free graph means $\rho = 0$, so at least 31 steps are required: $k \geq 31$.

The edge budget. Each step raises the edge count by one, so after k steps there are $66 + k$ edges. A cycle-free graph on 77 vertices is a forest, and by Lemma D it has at most 76 edges. So

$$66 + k \leq 76 \quad \implies \quad k \leq 10.$$

Again $31 \leq k \leq 10$ is impossible. $\square$

ANSWER

No, it is impossible. The cycle rank $m - n + c$ starts at $66 - 77 + 42 = 31$ and must reach 0; one step (delete one edge, add two) lowers it by at most 1, so at least 31 steps are needed. But each step adds a net edge, and a forest on 77 vertices has at most 76 edges, allowing at most 10 steps.

Machine check Lemma 19.1 was tested on 4000 random steps applied to random graphs on up to seven vertices: in every case the cycle rank dropped by at most one.

PROBLEM 20 $2k$ vertices, minimum degree $k - 1$ component sizes again

PROBLEM

Simple graph G has $2k$ vertices. Every vertex has degree at least $k - 1$, and there exists at least one vertex of degree k . Show that G is connected.

Theorem 20.1 *Let G be a simple graph on $n = 2k$ vertices with $\delta(G) \geq k - 1$, and suppose some vertex v_0 has $d(v_0) \geq k$. Then G is connected.*

Proof. Suppose G is disconnected, and let H_0 be the component containing v_0 and H_1 a different component. By Lemma B, a component containing a vertex of degree d has at least $d + 1$ vertices, so

$$|V(H_0)| \geq d(v_0) + 1 \geq k + 1, \quad |V(H_1)| \geq \delta(G) + 1 \geq k.$$

The two components are disjoint, so

$$2k = |V(G)| \geq |V(H_0)| + |V(H_1)| \geq (k + 1) + k = 2k + 1,$$

which is absurd. Hence G has a single component. □

ANSWER

G is connected. The component of the degree- k vertex holds at least $k + 1$ vertices and any other component at least $(k - 1) + 1 = k$; two of them would need $2k + 1$ vertices, one more than G has.

The extra vertex is indispensable Dropping the vertex of degree k makes the statement false: for $k = 3$, two disjoint triangles form a graph on $2k = 6$ vertices with every degree equal to $k - 1 = 2$, and it is disconnected. In general $K_k \dot{\cup} K_k$ is $(k - 1)$ -regular on $2k$ vertices and disconnected. The single vertex of degree k is exactly what breaks the tie $k + k = 2k$.

Machine check Exhaustive for $k = 1, 2, 3$ (that is, $n = 2, 4, 6$): among all graphs on those vertex sets, 12 037 satisfy the hypothesis and every one of them is connected. Without the degree- k requirement, 10 disconnected graphs on six vertices satisfy $\delta \geq 2$.

PROBLEM ★ Three colours, 123 edges

three spanning trees

PROBLEM

At most how many vertices may a graph G have, that has 123 edges and its edges can be coloured using colours red, white and green (every edge gets exactly one colour), so that between any two vertices of G there exist paths consisting of only red, only white as well as only green edges?

The condition says precisely that each of the three colour classes, taken as a spanning subgraph on all of $V(G)$, is connected.

Theorem ★.1 *Such a graph has at most 42 vertices, and 42 is attained.*

Upper bound. Let $n = |V(G)|$ and let R, W, Z be the sets of red, white and green edges; they partition $E(G)$, so $|R| + |W| + |Z| = 123$. The hypothesis says that any two vertices are joined by a red path, so the spanning subgraph $(V(G), R)$ is connected; by Theorem 17.1 it has at least $n - 1$ edges, i.e. $|R| \geq n - 1$, and likewise $|W| \geq n - 1$ and $|Z| \geq n - 1$. Adding,

$$123 = |R| + |W| + |Z| \geq 3(n - 1) \quad \implies \quad n - 1 \leq 41 \quad \implies \quad n \leq 42.$$

□

Attainment. For $n = 42$ equality must hold everywhere, so each colour class must be a connected spanning subgraph with exactly $41 = n - 1$ edges — that is, a spanning tree. Three pairwise edge-disjoint spanning trees on 42 vertices do exist. Work in $\mathbb{Z}_{42}$ and, for $k = 0, 1, 2$, let P_k be the zig-zag sequence

$$P_k: \quad k, \quad k + 1, \quad k - 1, \quad k + 2, \quad k - 2, \quad \dots, \quad k + 20, \quad k - 20, \quad k + 21 \pmod{42}.$$

Each P_k is a Hamiltonian path. Its terms are k , then $k + i$ and $k - i$ for $i = 1, \dots, 20$, and finally $k + 21$. That is $1 + 40 + 1 = 42$ terms, and they are pairwise distinct modulo 42: the residues $k, k \pm 1, \dots, k \pm 20$ are 41 distinct values and $k + 21 \equiv k - 21$ is the one remaining. So P_k visits every vertex exactly once and is a spanning tree with 41 edges.

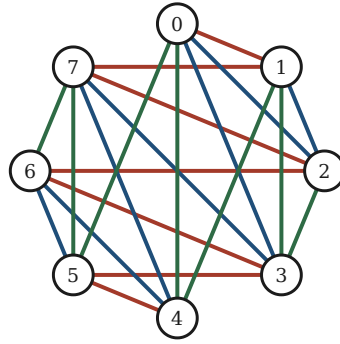
The three paths are pairwise edge-disjoint. Consecutive terms of P_k come in just two shapes, $\{k - (i - 1), k + i\}$ and $\{k + i, k - i\}$, whose endpoint sums are

$$(k - i + 1) + (k + i) = 2k + 1, \quad (k + i) + (k - i) = 2k \pmod{42}.$$

So every edge of P_k has endpoint sum $2k$ or $2k + 1$. For $k = 0, 1, 2$ these sum sets are $\{0, 1\}$, $\{2, 3\}$ and $\{4, 5\}$, which are pairwise disjoint; an edge belonging to two of the paths would have to have its endpoint sum in two disjoint sets. Hence no edge is shared.

Colouring P_0 red, P_1 white and P_2 green therefore gives a graph on 42 vertices with $3 \cdot 41 = 123$ edges in which every colour class is a spanning tree, hence connected.

□



$n = 8$ shown: three edge-disjoint spanning paths

The construction at $n = 8$: three edge-disjoint spanning paths in three colours. At $n = 42$ the same recipe gives $3 \times 41 = 123$ edges.

ANSWER

At most 42 vertices, and 42 is achievable. Each colour class must be a connected spanning subgraph, hence has at least $n - 1$ edges, so $3(n - 1) \leq 123$ and $n \leq 42$. At $n = 42$ each class is forced to be a spanning tree, and three edge-disjoint spanning trees on 42 vertices exist — for example three of the 21 Hamiltonian paths into which K_{42} decomposes.

Machine check The three zig-zag paths on 42 vertices were built explicitly and verified: each visits all 42 vertices exactly once, each has 41 edges and is connected, the endpoint sums of their edges are exactly $\{0, 1\}$, $\{2, 3\}$ and $\{4, 5\}$ as claimed, they are pairwise edge-disjoint, and their union has exactly 123 edges.

PROBLEM 21 Every graph has an acyclic orientation

order the vertices

PROBLEM

Prove that the edges of a simple graph $G = (V, E)$ can be oriented so that the directed graph obtained does not contain a directed cycle.

Theorem 21.1 Every finite simple graph admits an acyclic orientation.

Proof. Since V is finite, we may number its vertices $v_1, v_2, \dots, v_n$; write $\nu(v_i) = i$ for the label of a vertex. Orient every edge from its endvertex of smaller label to its endvertex of larger label:

$$\text{for } e = \{v_i, v_j\} \in E \text{ with } i < j, \text{ orient } e \text{ as } v_i \rightarrow v_j.$$

This is well defined: G is simple, so an edge has two distinct endvertices and their labels differ.

Suppose the resulting digraph contained a directed cycle $u_1 \rightarrow u_2 \rightarrow \dots \rightarrow u_k \rightarrow u_1$ with $k \geq 2$. Each arc goes from a smaller label to a larger one, so

$$\nu(u_1) < \nu(u_2) < \dots < \nu(u_k) < \nu(u_1),$$

where the last inequality comes from the closing arc $u_k \rightarrow u_1$. Chaining them gives $\nu(u_1) < \nu(u_1)$, which is false. Hence no directed cycle exists. $\square$

ANSWER

Number the vertices $1, \dots, n$ and orient each edge **from the smaller label to the larger**. Along any directed walk the labels strictly increase, so a directed cycle would return to its start with a strictly larger label than it left with — impossible. The orientation is therefore acyclic.

Remarks The proof uses only that V carries a linear order, so it works for any graph whose vertex set can be linearly ordered — every finite graph, and with the axiom of choice every infinite one too. Conversely, every acyclic orientation arises this way: an acyclic digraph has a topological order, and orienting by that order reproduces it. Simplicity matters only to exclude loops, which could never be oriented acyclically. Finally, the number of acyclic orientations of G is $|\chi_G(-1)|$, where χ_G is the chromatic polynomial (Stanley's theorem) — far more than we need, but it shows these orientations are plentiful, not exceptional.

Machine check Three thousand random simple graphs on up to seven vertices were oriented by the rule above; a depth-first search found a directed cycle in none of them.

PROBLEM 22 Three edges, three cuts

bridges after a deletion

PROBLEM

Let e, f and g be edges of simple, connected graph G . Assume that G remains connected if any one of its edges is deleted, however neither $G - e - g$ nor $G - f - g$ is connected. Prove that graph $G - f - g$ is also disconnected.

Reading the statement As printed, the conclusion " $G - f - g$ is disconnected" is already one of the hypotheses, so it cannot be what is meant. The hypotheses concern the pairs $\{e, g\}$ and $\{f, g\}$; the conclusion must therefore concern the third pair, $\{e, f\}$. We prove that $G - e - f$ is **disconnected**, and take e, f, g to be pairwise distinct, as the wording implies. (Some distinctness is genuinely needed: for $e = f$ the claim would read " $G - e$ is disconnected", contradicting the hypothesis that G survives any single deletion.)

Theorem 22.1 *Let G be connected with no bridge, and let e, f, g be three distinct edges such that $G - e - g$ and $G - f - g$ are both disconnected. Then $G - e - f$ is disconnected.*

Proof. Put $H := G - g$.

Step 1: H is connected, and both e and f are bridges of H . By hypothesis G stays connected after the deletion of any single edge, so $H = G - g$ is connected. Also $H - e = G - e - g$ is disconnected, so the edge e is a bridge of H ; likewise $H - f = G - f - g$ is disconnected, so f is a bridge of H .

Step 2: $H - e - f$ has exactly three components. Deleting the bridge e from the connected graph H leaves exactly two components (by Lemma C read backwards: e lies on no cycle of H , so $c(H) = c(H - e) - 1$, giving $c(H - e) = 2$). The edge $f \neq e$ survives in $H - e$ and lies

inside one of those two components; f is a bridge of H , hence also of $H - e$ — deleting edges cannot create a cycle through f — so deleting it splits that component in two. Therefore

$$c(H - e - f) = 3.$$

Step 3: putting g back. Now $G - e - f$ is exactly $H - e - f$ with the single edge g restored. By Lemma C one edge lowers the number of components by at most one, so

$$c(G - e - f) = c((H - e - f) + g) \geq 3 - 1 = 2.$$

A graph with at least two components is disconnected. □

ANSWER

$G - e - f$ is **disconnected**. Since G is bridgeless, $H = G - g$ is connected; and since $G - e - g$ and $G - f - g$ are disconnected, e and f are both bridges of H . Deleting two distinct bridges from a connected graph leaves three components, and restoring the single edge g can rejoin at most two of them — so $G - e - f$ has at least two components.

The same statement in cut language Because G is bridgeless while $G - e - g$ is disconnected, the pair $\{e, g\}$ is a *minimal* edge cut; likewise $\{f, g\}$. The cuts of a graph form a vector space over $\mathbb{F}_2$ under symmetric difference, and

$$\{e, g\} \triangle \{f, g\} = \{e, f\},$$

so $\{e, f\}$ is a cut too — which is exactly the conclusion. The elementary argument above avoids setting up the cut space, but this is what makes the result feel inevitable.

Machine check Every connected bridgeless graph on at most six vertices was enumerated together with every ordered triple of distinct edges (e, f, g) . In all 35 898 instances satisfying the hypotheses, $G - e - f$ was indeed disconnected.

PROBLEM 23

Deleting a column keeps the rows distinct

a graph on the rows

PROBLEM

All rows of an $n \times n$ table T are pairwise distinct. Prove that T has a column such that the table obtained by the deletion of that column still has pairwise distinct rows.

There is nothing graph-like in the statement, so we build the graph ourselves — on the rows — and then count edges against vertices using problem 17.

Definition Call column j *critical* if deleting it destroys distinctness, i.e. if some two rows of T agree in every column except j . The claim to prove is that not all n columns are critical.

Lemma 23.1 *If rows $r \neq s$ agree in every column except j , then j is determined by the pair $\{r, s\}$.*

Proof. Rows r and s are distinct, so they differ somewhere; agreeing outside j means they differ exactly in column j . If they also differed exactly in column j' with $j' \neq j$, then column j

would be one of the columns "outside j ", where they agree — contradicting that they differ in column j . $\square$

Theorem 23.2 *Some column of T may be deleted without making two rows equal.*

Proof. Suppose every one of the n columns is critical. Build a graph Γ :

- the n vertices of Γ are the n rows of T ;
- for each column j , choose one pair of rows $\{r_j, s_j\}$ agreeing outside column j (such a pair exists, as j is critical) and put the edge $r_j s_j$ into Γ , labelling it with j .

By Lemma 23.1 different columns yield different pairs, so these n edges are distinct: Γ has n vertices and n edges. By Theorem 17.2, Γ contains a cycle

$$r_1 \overset{j_1}{-} r_2 \overset{j_2}{-} \cdots \overset{j_{k-1}}{-} r_k \overset{j_k}{-} r_1, \quad k \geq 3,$$

whose edge labels $j_1, \dots, j_k$ are pairwise distinct columns.

Now look at what happens in column j_1 . The edge $r_1 r_2$ is labelled j_1 , so r_1 and r_2 *differ* in column j_1 . Every other edge of the cycle carries a label $j_i \neq j_1$, so its two rows *agree* in column j_1 . Walking the cycle the long way round,

$$r_2, r_3, \dots, r_k, r_1$$

are consecutive along edges labelled $j_2, \dots, j_k$, all different from j_1 ; hence all of these rows agree with one another in column j_1 . In particular r_2 and r_1 agree in column j_1 — contradicting that they differ there.

So not every column is critical, and any non-critical column may be deleted. $\square$

ANSWER

Call a column *critical* if deleting it makes two rows equal, and suppose all n were. Form a graph on the n rows by joining, for each column j , a pair of rows that agree everywhere except in column j ; different columns give different pairs, so this graph has n vertices and n edges and therefore contains a cycle. Along that cycle every edge but one keeps the entry in the first edge's column constant, forcing its two endpoints to agree in that column — while that edge says they differ. **Hence some column is not critical and can be deleted.**

Machine check All $n \times n$ tables with pairwise distinct rows were enumerated for $n = 2, 3$ over a two-letter alphabet and for $n = 3$ over a three-letter alphabet — 17 898 tables in total. Every one of them has a deletable column.

PROBLEM 24 Colouring K_4 in three colours*a decomposition***PROBLEM**

Is it possible to colour the edges of the complete graph K_4 using colours red, white and green in such a way that, however two of the colours are chosen of the three, the two graphs formed by the edges of these two colours, respectively, are isomorphic to each other?

The condition says that the three colour classes are pairwise isomorphic as spanning subgraphs of K_4 .

Theorem 24.1 *Such a colouring exists. Moreover every solution splits the six edges as $2 + 2 + 2$, and there are exactly two possibilities for the common shape of a colour class.*

Why $2 + 2 + 2$. Isomorphic graphs have the same number of edges, so the three classes have equal size; as they partition the $\binom{4}{2} = 6$ edges of K_4 , each has $6/3 = 2$ edges. A graph on four vertices with two edges is either two adjacent edges (a path P_3 plus an isolated vertex) or two disjoint edges (a perfect matching $2K_2$); these two are not isomorphic, having degree sequences $2, 1, 1, 0$ and $1, 1, 1, 1$. So all three classes are of the same one of these two shapes.

□

Two constructions. Take $V(K_4) = \{0, 1, 2, 3\}$.

Perfect matchings. Colour

$$\{01, 23\} \text{ red,} \quad \{02, 13\} \text{ white,} \quad \{03, 12\} \text{ green.}$$

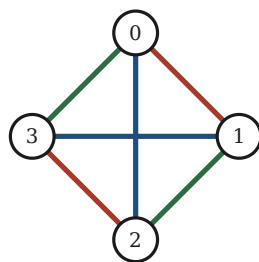
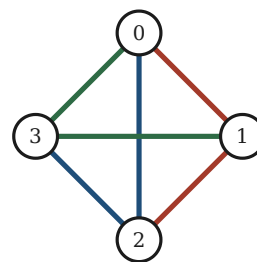
These are the three perfect matchings of K_4 ; they are disjoint and together use all six edges. Each class is a $2K_2$, so any two of them are isomorphic. (This is the proper 3-edge-colouring of the 3-regular graph K_4 .)

Paths. Colour

$$\{01, 12\} \text{ red,} \quad \{02, 23\} \text{ white,} \quad \{03, 13\} \text{ green.}$$

The six listed edges are again all of K_4 , and each class consists of two edges sharing a vertex (at 1, at 2, at 3 respectively), so each class is a P_3 plus an isolated vertex, and any two are isomorphic.

□

three perfect matchings ($2K_2$ each)three paths ($P_3 + K_1$ each)

The two kinds of solution. Left: the three perfect matchings. Right: three paths, centred at the vertices 1, 2 and 3.

ANSWER

Yes. The cleanest example is to give each of the three perfect matchings of K_4 its own colour: $\{01, 23\}$, $\{02, 13\}$, $\{03, 12\}$. Every colour class is then a pair of disjoint edges, so any two classes are isomorphic. A second, genuinely different solution colours three paths instead: $\{01, 12\}$, $\{02, 23\}$, $\{03, 13\}$.

How many such colourings are there? Exhaustive search over all $3^6 = 729$ colourings finds 54 that work. They fall into 9 unordered partitions, each of which can be assigned colours in $3! = 6$ ways: *one* partition into perfect matchings, and *eight* into paths. So the path-type solution is in fact the common one; the matching-type solution is unique.

PROBLEM 25 **Thirteen people***a Ramsey-type count***PROBLEM**

Prove that among any 13 people there exists one who knows at least 6 others, or there exist 3 people who are pairwise unknown to each other. (Acquaintance is symmetric.)

Model the party as a simple graph G on 13 vertices, joining two people when they know each other. The two alternatives read $\Delta(G) \geq 6$ and " G has three pairwise non-adjacent vertices".

Theorem 25.1 *Every simple graph on 13 vertices satisfies $\Delta(G) \geq 6$ or contains three pairwise non-adjacent vertices.*

Proof. Pick any vertex v and suppose $\Delta(G) \leq 5$, so in particular $d(v) \leq 5$. Let

$$S = V(G) \setminus (\{v\} \cup N(v))$$

be the set of people other than v whom v does *not* know. Since $|N(v)| = d(v) \leq 5$,

$$|S| \geq 13 - 1 - 5 = 7.$$

Case 1: two vertices of S are non-adjacent, say $x, y \in S$ with $xy \notin E(G)$. Neither x nor y is adjacent to v , by the definition of S . So v, x, y are three pairwise non-adjacent vertices — three mutual strangers.

Case 2: the vertices of S are pairwise adjacent. Then S induces a complete graph on $|S| \geq 7$ vertices, so each $x \in S$ is adjacent to the other $|S| - 1 \geq 6$ members of S , giving $d(x) \geq 6$ and hence $\Delta(G) \geq 6$, contrary to our supposition. So this case cannot occur under $\Delta(G) \leq 5$, and Case 1 applies.

Either way, one of the two alternatives holds. □

A second proof, in the complement

Suppose $\Delta(G) \leq 5$ and G has no three pairwise non-adjacent vertices. Pass to $\overline{G}$ on the same 13 vertices. "No three pairwise non-adjacent vertices in G " says $\overline{G}$ has no triangle, and $\Delta(G) \leq 5$ gives $\delta(\overline{G}) \geq 12 - 5 = 7$. But in a triangle-free graph adjacent vertices have no common neighbour, so for any edge uv of $\overline{G}$ the sets $N(u)$ and $N(v)$ are disjoint subsets of the 13 vertices:

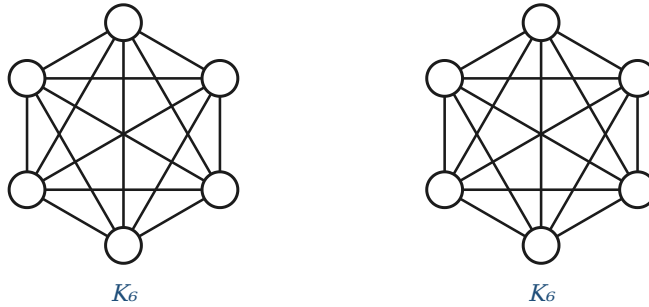
$$14 \leq d_{\overline{G}}(u) + d_{\overline{G}}(v) = |N(u) \cup N(v)| \leq 13,$$

a contradiction. (There is an edge to apply this to, since $\delta(\overline{G}) \geq 7 > 0$.)

ANSWER

Suppose nobody knows six others. Take any person v : they know at most 5, so at least $13 - 1 - 5 = 7$ people are strangers to v . If two of those seven are strangers to each other, they and v are **three mutual strangers**. Otherwise those seven all know one another, and then each of them knows at least 6 people — **somebody knows at least six**. Either way the claim holds.

Thirteen is sharp With twelve people the statement fails. Split them into two groups of six and let acquaintance mean "same group": each person then knows exactly 5 others, and any three people include two from the same group, who know each other. In graph terms $G = K_6 \dot{\cup} K_6$ is 5-regular on 12 vertices with no three pairwise non-adjacent vertices. So the count $13 - 1 - 5 = 7$ is exactly what makes the argument run.



12 people: everyone knows exactly 5, and any 3 contain 2 acquaintances

The twelve-person counterexample: two disjoint K_6 's. Everyone knows exactly five others, and no three people are pairwise strangers.

Machine check The triangle-free inequality $d(u) + d(v) \leq n$ was verified over all graphs on at most seven vertices; the 12-vertex example was checked to be 5-regular with no independent set of size 3; and 200 000 random graphs on 13 vertices with maximum degree ≤ 5 were searched — every one contains three pairwise non-adjacent vertices.

PROBLEM 26 Tiling a rectangle by integer-sided rectangles

double counting corners

PROBLEM

Show that if a rectangle T is tiled by rectangles so that each tile has at least one side of integer length, then T also has a side of integer length.

Place T with its lower-left corner at the origin, so $T = [0, w] \times [0, h]$, and call a point of the plane *integral* if both of its coordinates are integers. Everything follows from counting, in two ways, the pairs (tile, integral corner of that tile) — the same double count that proves the handshake lemma.

Lemma 26.1 *A rectangle with a side of integer length has an even number of integral corners.*

Proof. Let $R = [a, b] \times [c, d]$ and suppose its width $b - a$ is an integer (the case of integer height is identical with the roles of the axes exchanged). Then $a \in \mathbb{Z} \iff b \in \mathbb{Z}$. The four corners are $(a, c), (b, c), (a, d), (b, d)$, and

$$(a, c) \text{ integral} \iff a, c \in \mathbb{Z} \iff b, c \in \mathbb{Z} \iff (b, c) \text{ integral},$$

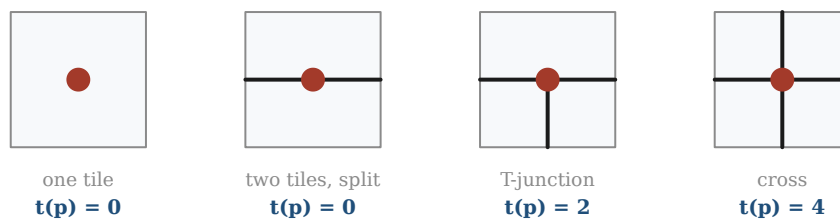
so the bottom two corners are integral together or not at all; likewise the top two. The count is thus 0, 2 or 4. $\square$

Lemma 26.2 *For a point p let $t(p)$ be the number of tiles having p as a corner. Then $t(p) = 1$ if p is a corner of T , and $t(p)$ is even for every other point p .*

Proof. Take $\delta > 0$ so small that the only tile boundaries meeting the δ -disc around p are those through p itself; then each of the four small quadrants at p that lies inside T is covered by a single tile. A tile A covering, say, the north-east quadrant has p as a corner exactly when it does not extend west or south of p — that is, exactly when A differs from the tiles covering the north-west and the south-east quadrants.

- p interior to T : all four quadrants are present. If one tile covers all four, $t(p) = 0$. If two tiles split the disc along a line through p , each contains p in the interior of a side and $t(p) = 0$. At a T-junction three tiles meet and exactly the two on the stem side have a corner at p , so $t(p) = 2$. If four distinct tiles meet, each has a corner at p and $t(p) = 4$. (Three quadrants cannot share one tile: an axis-parallel rectangle containing points of the north-east and south-west quadrants contains the whole box between them, hence all four.) In every case $t(p)$ is even.
- p on a side of T but not a corner: only two quadrants lie inside T , covered by tiles A and B . If $A = B$ then p is interior to a side of it and $t(p) = 0$; if $A \neq B$ then both have a corner at p and $t(p) = 2$.
- p a corner of T : one quadrant, one tile, and its corner is p . So $t(p) = 1$.

$\square$



at an interior point $t(p)$ is always even

The local pictures at an interior point. The number of tiles with a corner at p is 0, 2 or 4 — never odd.

Theorem 26.3 *If every tile has a side of integer length, then T has a side of integer length.*

Proof. Count the pairs (R, p) where R is a tile and p an integral corner of R ; call the total N .

By tiles. Every tile has an integer side, so by Lemma 26.1 contributes an even number. Hence N is even.

By points. Grouping instead by the integral point p ,

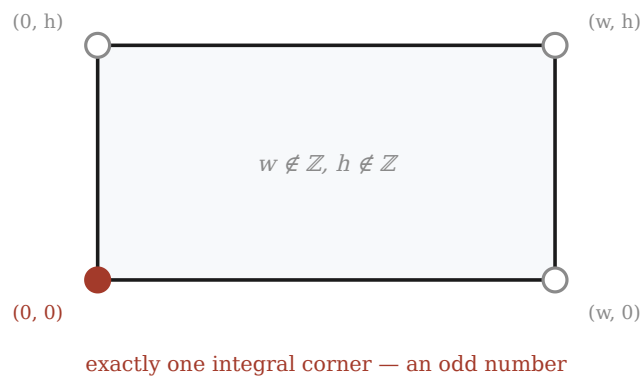
$$N = \sum_{p \text{ integral}} t(p).$$

By Lemma 26.2 every term is even except those coming from integral corners of T , each of which contributes exactly 1. Therefore

$$N \equiv \#\{\text{integral corners of } T\} \pmod{2}.$$

Since N is even, T has an *even* number of integral corners. But suppose neither w nor h were an integer. The corners of T are $(0, 0), (w, 0), (0, h), (w, h)$; the first is integral, while $(w, 0)$ and (w, h) fail because $w \notin \mathbb{Z}$ and $(0, h)$ fails because $h \notin \mathbb{Z}$. That is exactly *one* integral corner — an odd number, a contradiction.

Hence $w \in \mathbb{Z}$ or $h \in \mathbb{Z}$. □



If T had no integer side, only the origin among its four corners would be integral — and the double count forbids an odd number.

ANSWER

Put $T = [0, w] \times [0, h]$ and call a point integral when both coordinates are integers. Count the pairs (tile, integral corner of it) two ways. **By tiles:** a rectangle with an integer side has 0, 2 or 4 integral corners, so the total is even. **By points:** the number of tiles with a corner at a given point is even everywhere except at the four corners of T , where it is 1. So T must have an even number of integral corners — whereas a rectangle with neither side an integer has exactly one, namely the origin. **Hence w or h is an integer.**

Machine check Randomly generated tilings whose every tile has an integer side — 2 220 of them, with up to 33 tiles — were tested: every tile had an even number of integral corners, every container had an integer side, and across 5 737 corner points the count $t(p)$ was 1 at each corner of the container and even everywhere else, exactly as Lemma 26.2 asserts.

SUMMARY The nineteen answers

for reference

For convenience, the results proved above collected in one place. Problems 1–8 of the same sheet are solved in the companion volume.

#	problem	answer
9	All degrees distinct	only K_1
10	Degree sequence 2,2,2,4,5,5	no simple graph realises it
11	11 vertices, 45 edges	some degree ≥ 9
12	Cycle-avoiding game	first player wins $\Leftrightarrow n$ even
13	100 vertices, one of degree 66	connected
14	10 deletions, 33 additions	42 edges
15	5-vertex graphs, $m = 2 / 8 / 3$	2 / 2 / 4 graphs
16	Self-complementary	yes for $n = 5$, no for $n = 6$
17	Connectivity and cycles	$m \geq n-1$; $m = n \Rightarrow$ a cycle
18	Add 1 edge, delete 2	impossible ($41 > 24$)
19	Delete 1 edge, add 2	impossible ($31 > 10$)
20	$2k$ vertices, $\delta \geq k-1$	connected
★	123 edges, three colours	42 vertices
21	Acyclic orientation	order the vertices, orient upward
22	Three edges, three cuts	$G-e-f$ is disconnected
23	Deleting a column	always possible
24	Colouring K_4	yes — matchings, or paths
25	Thirteen people	true, and 13 is sharp
26	Tiling a rectangle	T has an integer side

Table B.1 — the nineteen answers at a glance.

Recurring techniques Four ideas do most of the work across these nineteen problems. *Counting degrees* settles 9, 10 and 11. *Counting components against edges* — the Edge Addition Lemma and its consequence that a forest on n vertices with c components has $n - c$ edges — settles 12, 14, 17, 18, 19, 22 and the starred problem, and supplies the cycle used in 23. *Bounding component sizes from below by degrees* settles 13 and 20. And *double counting a set of incidences* — the handshake lemma in disguise — settles 26. The remaining problems (15, 16, 21, 24, 25) are settled by an explicit construction together with an argument that no other kind exists.